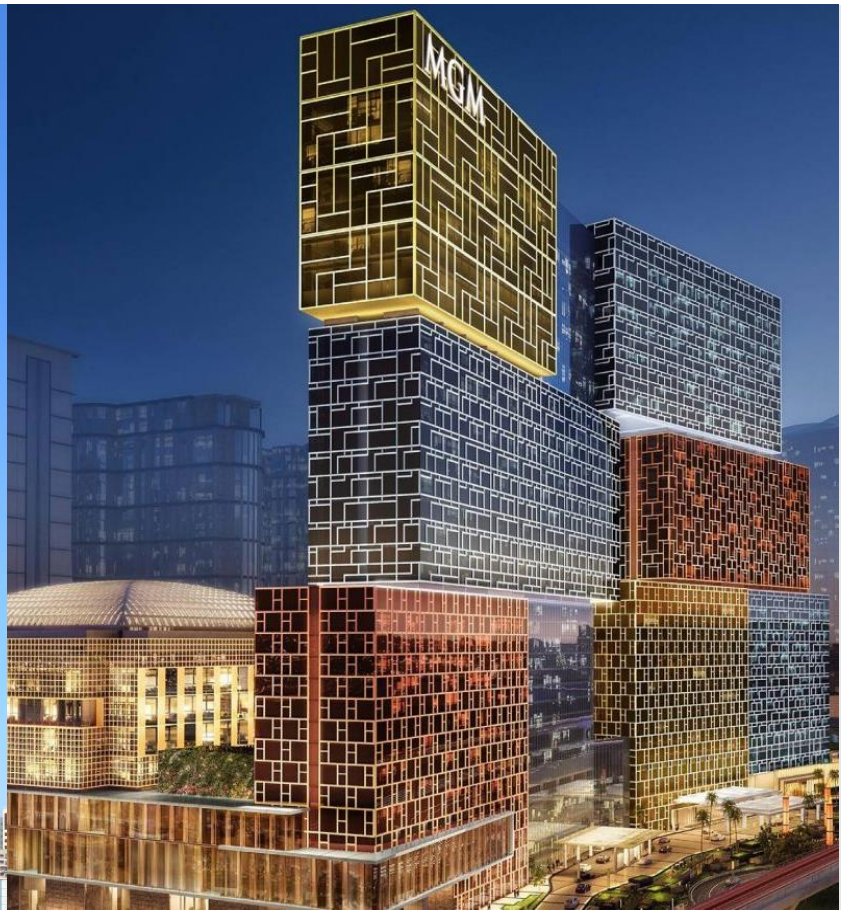
# MGM MACAU & COTAI
# RESOURCE EFFICIENCY MANAGEMENT REPORT
FEBRUARY 2024

Prepared For:

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# EXECUTIVE SUMMARY

NV5 has been working closely with MGM to implement the EMOs as well as identify additional opportunities to achieve ISO certification energy targets. NV5 appreciates the great support and trust from MGM team over the years, in particular the Facility Management and the Sustainability Teams.

## Key Performance Indicators

NV5 reports both normalized and actual savings on a monthly basis for both properties. The normalized savings are determined based on a 2019 baseline, with aligned annual savings targets to meet both ISO50001 as well as NV5's retainer contract requirements. Please note that the reported normalized savings achieved are to-date and cumulative for each month of the respective year.

*Table 1. ISO and Retainer Contract Energy Performance Targets for 2025*

| MGM Properties | Baseline | Normalized Savings | |
|---|---|---|---|
| MACAU | 2019 | 2025 Target | **-13%** |
| | | Savings as of Feb 2024 | **-19.6%** |
| COTAI | 2019 | 2025 Target | **-13%** |
| | | Savings as of Feb 2024 | **-7.4%** |

The actual monthly savings are determined against the 2019 baseline consumption and cumulative for each month of 2024.

*Table 2. Internal Actual Energy Performance Targets for 2025*

| MGM Properties | Baseline | Actual Saving Internal Target[1] | Savings Achieved Up to February 2024 |
|---|---|---|---|
| MACAU | 2019 | 13% | **+7.2%** |
| COTAI | 2019 | 13% | **-1.7%** |

[1] Actual Energy Savings Target for 2025 has been established based on future CAPEX projects, following review with MGM teams.

## Highlighted Changes in Operation

**MACAU**

1. MGM MACAU completed the CT-10 fan & motor retrofit pilot project on Jan 2024. Measurement & Verification (M&V) works were conducted to verify the new CT fan and motor performance on Feb 2024.
2. MGM MACAU completed the Natural Gas changeover on 16 Jan 2024.
3. MGM MACAU switched ON three boilers (1 big boiler and 2 small boilers) during 1 - 29 Feb 2024 due to low OAT fluctuation and final testing for NG system.
4. MGM MACAU completed the rooftop façade LED replacement on Feb 2024.
5. MGM MACAU will replace the Lion Backdrop lighting by Q2 2024.

**COTAI**

1. MGM COTAI completed CDP Extension Work at B2 Carpark (Phase 11) and (Phase 12) is in progress.
2. MGM COTAI further adjusted the back of house AHUs for night mode low fan speed operation.
3. MGM COTAI continues to replace the back of house office lighting panel with occupancy sensor in different phases.
4. MGM COTAI has resumed the power supply of MDM Kitchen and Restaurant to their original circuits.

### Consumption Intensity

The 2024 YTD electricity and water use intensities are tracked on a monthly basis for both properties:

*Table 3. Year to Date Actual Electricity Use Intensity (EUI)*

| MGM Properties | YTD EUI (kWh/m²) | % Change Compared to 2023 |
|---|---|---|
| MACAU | **23.3** | **+6.8%** |
| COTAI | **23.6** | **+8.0%** |

*Table 4. Year to Date Actual Water Use Intensity (WUI)*

| MGM Properties | YTD WUI (m³/ m²) | % Change Compared to 2023 |
|---|---|---|
| MACAU | **0.25** | **+24.3%** |
| COTAI | **0.18** | **+13.7%** |

## ISO 50001 Certification Audit

### MACAU

The HKQAA 2024 renewal audit was completed on 1 March 2024. There is no nonconformity (NC) and areas for improvement (AFI) from auditors.

### COTAI

The HKQAA 2023 surveillance audit was completed on 7 December 2023. There is no nonconformity (NC) and areas for improvement (AFI) from auditors.

## Milestones in Year 2024

The following activities are to be completed in year 2024:

*Table 5. MGM MACAU Year 2024 Projects*

| Item | Initiatives | Status |
|---|---|---|
| **Electricity** | | |
| 1 | Chiller Replacement (Phase 1 : 1 unit) | On-going |
| 2 | Cooling Tower Fan Retrofit (Phase 1 : 1 unit) | Completed |
| 3 | Cooling Tower Fan Retrofit (Phase 2 : 7 units) | On-going |
| 4 | Secondary Chilled Water Pump Retrofit (Phase 1 : 3 units) | On-going |
| 5 | AHU EC Fan Replacement (Phase 1) | On-going |
| 6 | Neon Light LED Replacement (Phase 3) | On-going |
| 7 | Rooftop Façade LED Replacement | Completed |
| 8 | Lion Backdrop Lighting Upgrade | On-going |
| **LPG** | | |
| 9 | Natural Gas Changeover Project | Completed |

*Table 6. MGM COTAI Year 2024 Projects*

| Item | Initiatives | Status |
|---|---|---|
| **Electricity** | | |
| 1 | Chiller Plant Controls Optimization<br>(CHWP DP Setpoint Reset, CHWST Setpoint Reset…) | On-going |
| 2 | PICCV Implementation 2024 (14 Units) | On-going |
| 3 | AHU EC Fan Replacement (Phase 1) | On-going |
| 4 | Additional VSD for Exhaust Air Fan (Phase 2 : 12 units) | On-going |
| 5 | KEF Steps Speed Control Optimization 2024 (Hu Dao, Coast, Grill 58, Mansion) | On-going |
| 6 | Back of House Offices Lighting Replacement | On-going |
| **Water** | | |
| 7 | Extension of B2 Carpark CDP Water Recycling (Phase 11) | Completed |
| 8 | Extension of B2 Carpark CDP Water Recycling (Phase 12) | On-going |

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# ACKNOWLEDGMENT

NV5 has been working with MGM Team as the energy and sustainability manager to achieve energy savings in MGM properties and feed operational intelligence into the design.

The major goal of this project is not only to realize operation energy savings but also to extend the life of equipment, hence reducing CAPEX. MGM has demonstrated continuous commitment to be an environmentally and socially responsible business in the past years. In addition to that NV5 has assisted MGM MACAU and COTAI to achieve several other energy management and sustainability awards and certifications, such as ISO 50001, CEM Energy Savings Award, Macao Green Hotel Award and China 3 Star Green Building Operation Label.

NV5 appreciates the great support and trust from MGM Team over the years, especially the Facilities Management and the Sustainability Teams.

# 1.0 PORTFOLIO PERFORMANCE SUMMARY

## 1.1 ELECTRICITY USE INTENSITY (EUI)

MACAU's electricity use intensity (EUI) in February 2024 had increased by 8.3% compared to the same month in 2023 and increased by 6.8% in 2024 compared to 2023. Meanwhile, COTAI's EUI in February 2024 had increased by 9.2% compared to the same month in 2023 and increased by 8.0% in 2024 compared to 2023.

*Table 7. Actual Electricity Consumption Intensity (EUI) Comparison*

| Monthly Period | MACAU | | | COTAI | | |
|---|---|---|---|---|---|---|
| | Year 2023 Actual EUI (kWh/m²) | Year 2024 Actual EUI (kWh/m²) | % Change | Year 2023 Actual EUI (kWh/m²) | Year 2024 Actual EUI (kWh/m²) | % Change |
| **January** | 22.4 | 23.7 | +5.4% | 22.6 | 24.2 | +6.9% |
| **February** | 21.2 | 23.0 | +8.3% | 21.0 | 23.0 | +9.2% |
| **March** | | | | | | |
| **April** | | | | | | |
| **May** | | | | | | |
| **June** | | | | | | |
| **July** | | | | | | |
| **August** | | | | | | |
| **September** | | | | | | |
| **October** | | | | | | |
| **November** | | | | | | |
| **December** | | | | | | |
| **Average** | **21.8** | **23.3** | **+6.8%** | **21.8** | **23.6** | **+8.0%** |

## 1.2 WATER USE INTENSITY (WUI)

MACAU's water use intensity (WUI) in February 2024 has increased by **20.4%** compared to the same month in 2023 and increased by **24.3%** in 2024 compared to 2023. Meanwhile, COTAI's WUI in February 2024 has increased by **9.1%** compared to the same month in 2023 and increased by **13.7%** in 2024 compared to 2023.

*Table 8. Actual Water Consumption Intensity (WUI) Comparison*

| Monthly Period | MACAU | | | COTAI | | |
|---|---|---|---|---|---|---|
| | Year 2023 Actual WUI ($m^3/m^2$) | Year 2024 Actual WUI ($m^3/m^2$) | % Change | Year 2023 Actual WUI ($m^3/m^2$) | Year 2024 Actual WUI ($m^3/m^2$) | % Change |
| **January** | 0.19 | 0.24 | +28.7% | 0.14 | 0.17 | +19.0% |
| **February** | 0.21 | 0.26 | +20.4% | 0.16 | 0.18 | +9.1% |
| **March** | | | | | | |
| **April** | | | | | | |
| **May** | | | | | | |
| **June** | | | | | | |
| **July** | | | | | | |
| **August** | | | | | | |
| **September** | | | | | | |
| **October** | | | | | | |
| **November** | | | | | | |
| **December** | | | | | | |
| **Average** | **0.20** | **0.25** | **+24.3%** | **0.15** | **0.18** | **+13.7%** |

# 2.0 MGM MACAU PERFORMANCE

## 2.1 ELECTRICITY CONSUMPTION

The actual electricity consumption and cost in February 2024 increased by 8.3% and 8.3% respectively compared to the same month in 2023. To date, MGM MACAU has increased the actual electricity consumption by 6.8% and associated costs increased by 6.8%.

*Table 9. MACAU Summary of Actual Monthly Electricity Consumption and Cost*

| Monthly Period | Actual Electricity Consumption (kWh) | | | Actual Electricity Cost (MOP) | | | % Change – CDD |
|---|---|---|---|---|---|---|---|
| | 2023 | 2024 | % Change Actual | 2023 | 2024 | % Change Actual | |
| January | 4,745,400 | 5,003,520 | +5.4% | 5,668,020 | 5,973,270 | +5.4% | +173.7% |
| February | 4,488,360 | 4,860,480 | +8.3% | 5,365,140 | 5,812,030 | +8.3% | +17.2% |
| March | | | | | | | |
| April | | | | | | | |
| May | | | | | | | |
| June | | | | | | | |
| July | | | | | | | |
| August | | | | | | | |
| September | | | | | | | |
| October | | | | | | | |
| November | | | | | | | |
| December | | | | | | | |
| **Total** | **9,233,760** | **9,864,000** | **+6.8%** | **11,033,160** | **11,785,300** | **+6.8%** | **+62.7%** |

***Important note**: Cooling Degree Days (CDDs) and food cover are the main variables taken into consideration on electricity consumption normalization.*

NV5 summarized the key items that could result in an electricity consumption increase below:

- Chiller Plant electricity consumption has increased by 30.5% in February 2024 compared with the same month in 2023. The cooling degree days have increased by 17.2% respectively in February 2024.

Table 10 presents the comparison of electricity rates in the years 2023 and 2024. The electricity rate in February 2024 increased by 0.04% compared to the same month in 2023.

*Table 10. MACAU - Summary of Actual Monthly Electricity Rate*

| Monthly Period | Year 2023 – Electricity Rate (MOP/kWh) | Year 2024 – Electricity Rate (MOP/kWh) | % Change – Electricity Rate |
|---|---|---|---|
| **January** | 1.1944 | 1.1938 | -0.05% |
| **February** | 1.1953 | 1.1958 | +0.04% |
| **March** | | | |
| **April** | | | |
| **May** | | | |
| **June** | | | |
| **July** | | | |
| **August** | | | |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Average** | **1.1949** | **1.1948** | **-0.01%** |

Table 11 summarizes the monthly normalized electricity consumption. The actual electricity consumption in February 2024 decreased by 23.5% compared to 2019 normalized consumption in the same month.

*Table 11. MACAU - Summary of Normalized Monthly Electricity Consumption*

| Monthly Period | 2019 Normalized Electricity Consumption (kWh) | 2024 Actual Electricity Consumption (kWh) | % Change – Normalized |
|---|---|---|---|
| **January** | 6,303,975 | 5,003,520 | -20.6% |
| **February** | 6,353,462 | 4,860,480 | -23.5% |
| **March** | | | |
| **April** | | | |
| **May** | | | |
| **June** | | | |
| **July** | | | |
| **August** | | | |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total** | **12,657,437** | **9,864,000** | **-22.1%** |

## 2.2 GAS CONSUMPTION

Gas consumption can be broken down into three (3) categories: boiler, hotel upper zone DHW, and kitchens/restaurants. Since the boiler and restaurants Gas utilization cover over 95% of the overall site Gas consumption, NV5 focuses the analysis on those two areas.

As of 16 January 2024, MGM Macau has transitioned from LPG to Natural Gas for all gas-operated equipment.

According to Table 12, the actual Gas consumption in February 2024 increased by **54.3%** while the cost increased by **37.0%** respectively compared to the same month in 2023. To date, MGM MACAU has increased the actual Gas consumption by **23.2%** and associated costs increased by **22.2%.**

*Table 12. MACAU – Summary of Actual Monthly Gas Consumption and Cost*

| Monthly Period | Actual Gas Consumption (GJ) | | | Actual Gas Cost (MOP) | | | % Change – Min OAT |
|---|---|---|---|---|---|---|---|
| | 2023 (LPG) | 2024 (LPG + NG) | % Change Actual | 2023 (LPG) | 2024 (LPG + NG) | % Change Actual | |
| **January** | 5,577 | 5,760 | +3.3% | 1,084,579 | 1,211,135 | +11.7% | -37.6% |
| **February** | 3,555 | 5,487 | +54.3% | 768,347 | 1,052,400 | +37.0% | -35.8% |
| **March** | | | | | | | |
| **April** | | | | | | | |
| **May** | | | | | | | |
| **June** | | | | | | | |
| **July** | | | | | | | |
| **August** | | | | | | | |
| **September** | | | | | | | |
| **October** | | | | | | | |
| **November** | | | | | | | |
| **December** | | | | | | | |
| **Total** | **9,132** | **11,247** | **+23.2%** | **1,852,926** | **2,263,535** | **+22.2 %** | **-36.5%** |

***Important note:*** *Minimum Outdoor Air Temperature (OAT) is the main variable taken into consideration for Gas consumption normalization.*

NV5 summarized the key items which could result in Gas consumption increase below:

- Kitchen gas consumption has increased 11.0% due to Food Cover increased by 66.8% in February 2024 compared with the same month in 2023.
- There was 10 days under low OAT conditions (below 15°C) in February 2024 that Large Boiler was operated to support daily operation needs. Boiler gas consumption has increased by 37.2% in February 2024 compared with the same month in 2023.

Table 13 presents the comparison of Gas rates in the years 2023 and 2024. The Gas rates in February 2024 decreased by 11.2% compared with the same month in 2023.

*Table 13. MACAU - Summary of Actual Monthly Gas Rate*

| Monthly Period | Year 2023 – Gas Rate (MOP/GJ) | Year 2024 – Gas Rate (MOP/GJ) | % Change – Gas Rate |
|---|---|---|---|
| **January** | 194.48 | 210.27 | +8.1% |
| **February** | 216.11 | 191.80 | -11.2% |
| **March** | | | |
| **April** | | | |
| **May** | | | |
| **June** | | | |
| **July** | | | |
| **August** | | | |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Average** | **205.30** | **201.04** | **-2.1%** |

Table 14 summarizes the monthly normalized Gas consumption. The actual Gas consumption in February 2024 has decreased by 9.0% compared to 2019 normalized consumption in the same month.

*Table 14. MACAU - Summary of Normalized Monthly Gas Consumption*

| Monthly Period | 2019 Normalized Gas Consumption (GJ) | 2024 Actual Gas Consumption (GJ) | % Change – Normalized |
|---|---|---|---|
| **January** | 6,538 | 5,760 | -11.9% |
| **February** | 6,032 | 5,487 | -9.0% |
| **March** | | | |
| **April** | | | |
| **May** | | | |
| **June** | | | |
| **July** | | | |
| **August** | | | |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total** | **12,570** | **11,247** | **-10.5%** |

## 2.3 ISO50001 ENERGY CONSUMPTION RECORD

The MACAU ISO50001 energy performance target is to achieve 13% normalized energy savings by 2025 compared to the 2019 baseline. To date, normalized energy saving achieved up to February 2024 is **19.6%** as shown in Table 15. Table 16 below summarizes the site monthly energy consumption (Electricity and LPG/NG combined) compared to the 2019 baseline.

*Table 15. MACAU ISO 50001 Normalized Energy Performance Targets for 2025*

| MGM Properties | Baseline | Normalized Savings | |
|---|---|---|---|
| MACAU | 2019 | 2025 Target | **-13%** |
| | | Savings as of Feb 2024 | **-19.6%** |

*Table 16. MACAU ISO 50001 Normalized Monthly Energy Consumption*

| Monthly Period | 2019 Normalized Energy Consumption (GJ) | 2024 Actual Energy Consumption (GJ) | % Change – Normalized |
|---|---|---|---|
| **January** | 29,232 | 23,773 | -18.7% |
| **February** | 28,904 | 22,985 | -20.5% |
| **March** | | | |
| **April** | | | |
| **May** | | | |
| **June** | | | |
| **July** | | | |
| **August** | | | |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total** | **58,137** | **46,757** | **-19.6%** |

***Important note:*** *Factors used for unit conversion: Electricity (0.0036), LPG (0.0453), and NG (0.0347)*

## 2.4 WATER CONSUMPTION

Based on the existing water metering system, the overall site water consumption can be broken down into five (5) categories: Kitchen and Restaurants; Hotel Tower; Fire Services; Chiller Plant & miscellaneous.

According to Table 17, the actual water consumption and cost in February 2024 has increased by 20.4% and 20.4% respectively compared to the same month in 2023 respectively. To date, MGM MACAU has increased both the actual water consumption and associated cost by 24.3% and 24.2% respectively.

*Table 17. MACAU - Summary of Actual Monthly Water Consumption and Cost*

| Monthly Period | Actual Water Consumption ($m^3$) | | | Actual Water Cost (MOP) | | | Normalization Factors | |
|---|---|---|---|---|---|---|---|---|
| | 2023 | 2024 | % Change Actual | 2023 | 2024 | % Change Actual Cost | % Change – CDD | % Change – Food Covers |
| **January** | 39,305 | 50,588 | +28.7% | 305,020 | 392,330 | +28.6% | +173.7% | +50.5% |
| **February** | 45,112 | 54,323 | +20.4% | 350,080 | 421,370 | +20.4% | +17.2% | +66.8% |
| **March** | | | | | | | | |
| **April** | | | | | | | | |
| **May** | | | | | | | | |
| **June** | | | | | | | | |
| **July** | | | | | | | | |
| **August** | | | | | | | | |
| **September** | | | | | | | | |
| **October** | | | | | | | | |
| **November** | | | | | | | | |
| **December** | | | | | | | | |
| **Total** | **84,417** | **104,911** | **+24.3%** | **655,100** | **813,700** | **+24.2%** | **+62.7%** | **+58.3%** |

***Important note:*** *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on water consumption normalization.*

NV5 summarized the key items which could result in water consumption increase below:

- Kitchen water consumption has increased by 39.6% due to Food Cover increased by 66.8% in February 2024 compared with the same month in 2023.
- Podium water consumption for February 2024 has increased by 27.9% compared with the same month in 2023.

Table 18 presents the comparison of water rates in the years 2023 and 2024. The water rate in February 2024 has decreased by 0.04% compared to the same month in 2023. The average utility rate of water in 2024 has decreased by 0.05% compared to 2023.

*Table 18. MACAU - Summary of Actual Monthly Water Rates*

| Monthly Period | Year 2023 – Water Rate (MOP/m³) | Year 2024 – Water Rate (MOP/m³) | % Change – Water Rate |
|---|---|---|---|
| **January** | 7.760 | 7.755 | -0.06% |
| **February** | 7.760 | 7.757 | -0.04% |
| **March** | | | |
| **April** | | | |
| **May** | | | |
| **June** | | | |
| **July** | | | |
| **August** | | | |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Average** | **7.760** | **7.756** | **-0.05%** |

Table 19 summarized the monthly normalized water consumption. The actual water consumption in February 2024 has increased by 15.8% compared to 2019 normalized consumption in the same month.

*Table 19. MACAU - Summary of Normalized Monthly Water Consumption*

| Monthly Period | 2019 Normalized Water Consumption (m³) | 2024 Actual Water Consumption (m³) | % Change – Normalized |
|---|---|---|---|
| **January** | 57,198 | 50,588 | -11.6% |
| **February** | 46,907 | 54,323 | +15.8% |
| **March** | | | |
| **April** | | | |
| **May** | | | |
| **June** | | | |
| **July** | | | |
| **August** | | | |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total** | **104,104** | **104,911** | **+0.8%** |

## 2.5 MACAU YEAR 2024 PROJECTS

NV5 will continue to perform analysis and coordinate with MGM to implement the following projects in 2024 and reduce the site energy and water consumption which supports MGM energy and water savings target.

*Table 20. MACAU Year 2024 Projects*

| Item | Initiatives | Status |
|---|---|---|
| **Electricity** | | |
| 1 | Chiller Replacement (Phase 1 : 1 unit) | On-going |
| 2 | Cooling Tower Fan Retrofit (Phase 1 : 1 unit) | Completed |
| 3 | Cooling Tower Fan Retrofit (Phase 2 : 7 units) | On-going |
| 4 | Secondary Chilled Water Pump Retrofit (Phase 1 : 3 units) | On-going |
| 5 | AHU EC Fan Replacement (Phase 1 : 5 units) | On-going |
| 6 | Neon Light LED Replacement (Phase 3) | On-going |
| 7 | Rooftop Façade LED Replacement | Completed |
| 8 | Lion Backdrop Lighting Upgrade | On-going |
| **LPG** | | |
| 9 | Natural Gas Changeover Project | Completed |

# 3.0 MGM COTAI PERFORMANCE

## 3.1 ELECTRICITY CONSUMPTION

The actual electricity consumption and cost in February 2024 increased by 9.2% and 9.2% respectively compared to the same month in 2023. To date, MGM COTAI has increased the actual electricity consumption by 8.0% and associated cost increased by 8.0%.

*Table 21. COTAI Summary of Actual Monthly Electricity Consumption and Cost*

| Monthly Period | Actual Electricity Consumption (kWh) | | | Actual Electricity Cost (MOP) | | | % Change – CDD |
|---|---|---|---|---|---|---|---|
| | 2023 | 2024 | % Change Actual | 2023 | 2024 | % Change Actual | |
| January | 8,471,280 | 9,054,720 | +6.9% | 10,516,100 | 11,239,000 | +6.9% | +173.7% |
| February | 7,869,600 | 8,597,400 | +9.2% | 9,794,150 | 10,692,600 | +9.2% | +17.2% |
| March | | | | | | | |
| April | | | | | | | |
| May | | | | | | | |
| June | | | | | | | |
| July | | | | | | | |
| August | | | | | | | |
| September | | | | | | | |
| October | | | | | | | |
| November | | | | | | | |
| December | | | | | | | |
| **Total** | **16,340,880** | **17,652,120** | **+8.0%** | **20,310,250** | **21,931,600** | **+8.0%** | **+62.7%** |

***Important note**: Cooling Degree Days (CDDs), Food Cover, Room Night and E-Bus Mileage are the main variables taken into consideration on electricity consumption normalization.*

NV5 summarized the key items which could result in electricity consumption increase below:

- Chiller Plant electricity consumption has increased by 21.9% in February 2024 compared with the same month in 2023. The cooling degree days have increased by 17.2% respectively in February 2024.
- EV Charger (Guest) and E-Shuttle Bus electricity consumption for February 2024 has increased by 216.5% and 165.6% compared to the same month in 2023.
- There was 10 days under low OAT conditions (below 15°C) in February 2024. Heat Pump Plant electricity consumption for February 2024 has increased by 23.4% compared to the same month in 2023.
- Kitchen & restaurant electricity consumption for February 2024 has increased by 11.1% compared to the same month in 2023.

Table 22 presents the monthly e-bus electricity consumption in MGM COTAI. The e-bus electricity consumption in February 2024 has increased by **165.6%** compared to the same month in 2023.

*Table 22. COTAI - Summary of Monthly e-bus Electricity Consumption*

| Monthly Period | E-bus Electricity Consumption (kWh) | | |
|---|---|---|---|
| | 2023 | 2024 | % Change – Actual |
| **January** | 51,261 | 137,030 | +167.3% |
| **February** | 53,640 | 142,468 | +165.6% |
| **March** | | | |
| **April** | | | |
| **May** | | | |
| **June** | | | |
| **July** | | | |
| **August** | | | |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total** | **104,901** | **279,498** | **+166.4%** |

Table 23 presents the comparison of electricity rates in the years 2023 and 2024. The electricity rate in February 2024 decreased by **0.07%** compared to the same month in 2023.

*Table 23. COTAI - Summary of Actual Monthly Electricity Rate*

| Monthly Period | Year 2023 – Electricity Rate (MOP/kWh) | Year 2024 – Electricity Rate (MOP/kWh) | % Change – Electricity Rate |
|---|---|---|---|
| **January** | 1.2414 | 1.2412 | -0.01% |
| **February** | 1.2446 | 1.2437 | -0.07% |
| **March** | | | |
| **April** | | | |
| **May** | | | |
| **June** | | | |
| **July** | | | |
| **August** | | | |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Average** | **1.2430** | **1.2425** | **-0.04%** |

Table 24 summarizes the monthly normalized electricity consumption. The actual electricity consumption in February 2024 decreased by 5.1% compared to 2019 normalized consumption in the same month.

*Table 24. COTAI - Summary of Normalized Monthly Electricity Consumption*

| Monthly Period | 2019 Normalized Electricity Consumption (kWh) | 2024 Actual Electricity Consumption (kWh) | % Change – Normalized |
|---|---|---|---|
| **January** | 9,166,278 | 9,054,720 | -1.2% |
| **February** | 9,055,928 | 8,597,400 | -5.1% |
| **March** | | | |
| **April** | | | |
| **May** | | | |
| **June** | | | |
| **July** | | | |
| **August** | | | |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total** | **18,222,206** | **17,652,120** | **-3.1%** |

## 3.2 NATURAL GAS CONSUMPTION

The actual natural gas consumption and cost in February 2024 increased by 29.0% and 31.6% respectively compared to the same month in 2023, respectively. To date, MGM COTAI has increased actual natural gas consumption by 33.6% and associated costs by 41.0%.

*Table 25. COTAI - Summary of Actual Monthly Natural Gas Consumption and Cost*

| Monthly Period | Actual NG Consumption ($m^3$) | | | Actual NG Cost (MOP) | | | % Change – Min OAT | % Change – Food Covers |
|---|---|---|---|---|---|---|---|---|
| | 2023 | 2024 | % Change Actual | 2023 | 2024 | % Change Actual | | |
| **January** | 64,043 | 88,198 | +37.7% | 405,800 | 607,700 | +49.8% | -37.6% | +49.1% |
| **February** | 58,053 | 74,893 | +29.0% | 378,840 | 498,460 | +31.6% | -35.8% | +48.7% |
| **March** | | | | | | | | |
| **April** | | | | | | | | |
| **May** | | | | | | | | |
| **June** | | | | | | | | |
| **July** | | | | | | | | |
| **August** | | | | | | | | |
| **September** | | | | | | | | |
| **October** | | | | | | | | |
| **November** | | | | | | | | |
| **December** | | | | | | | | |
| **Total** | **122,096** | **163,091** | **+33.6%** | **784,640** | **1,106,160** | **+41.0%** | **-36.5%** | **+48.9%** |

***Important note:*** *Minimum Outdoor Air Temperature is the main variable taken into consideration on natural gas consumption normalization*

NV5 summarized the key items which could result in natural gas consumption increase below:

- Kitchen NG consumption for February 2024 has increased by 27.5% compared to the same month in 2023. Food cover has increased by 48.7% in February 2024.

Table 26 presents the comparison of natural gas rates in the years 2023 and 2024. The natural gas rate in February 2024 increased by 2.0% compared to the same month in 2023.

*Table 26. COTAI - Summary of Actual Monthly Natural Gas Rate*

| Monthly Period | Year 2023 – Natural Gas Rate (MOP/m³) | Year 2024 – Natural Gas Rate (MOP/m³) | % Change – NG Rate |
|---|---|---|---|
| **January** | 6.34 | 6.89 | +8.7% |
| **February** | 6.53 | 6.66 | +2.0% |
| **March** | | | |
| **April** | | | |
| **May** | | | |
| **June** | | | |
| **July** | | | |
| **August** | | | |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Average** | **6.43** | **6.77** | **+5.3%** |

Table 27 summarizes the monthly normalized natural gas consumption. The actual natural gas consumption in February 2024 decreased by 42.0% compared to 2019 normalized consumption in the same month.

*Table 27. COTAI - Summary of Normalized Monthly Natural Gas Consumption*

| Monthly Period | 2019 Normalized Natural Gas Consumption (m³) | 2024 Actual Natural Gas Consumption (m³) | % Change – Normalized |
|---|---|---|---|
| **January** | 134,135 | 88,198 | -34.2% |
| **February** | 129,115 | 74,893 | -42.0% |
| **March** | | | |
| **April** | | | |
| **May** | | | |
| **June** | | | |
| **July** | | | |
| **August** | | | |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total** | **263,250** | **163,091** | **-38.0%** |

## 3.3 ISO50001 ENERGY CONSUMPTION RECORD

COTAI ISO50001 energy target is to achieve 13% normalized energy consumption reduction by 2025 compared to the 2019 baseline. To date, normalized energy saving achieved up to February 2024 is **7.4%** as shown in Table 28 and Table 29 below summarizes the site monthly energy consumption (Electricity and Natural Gas combined) compared to the 2019 baseline.

*Table 28. COTAI ISO 50001 Normalized Energy Performance Targets for 2025*

| MGM Properties | Baseline | Normalized Savings | |
|---|---|---|---|
| COTAI | 2019 | 2025 Target | **-13%** |
| | | Savings as of Feb 2024 | **-7.4%** |

*Table 29. COTAI ISO 50001 Normalized Monthly Energy Consumption*

| Monthly Period | 2019 Normalized Energy Consumption (GJ) | 2024 Actual Energy Consumption (GJ) | % Change – Normalized |
|---|---|---|---|
| **January** | 37,653 | 35,657 | -5.3% |
| **February** | 37,082 | 33,549 | -9.5% |
| **March** | | | |
| **April** | | | |
| **May** | | | |
| **June** | | | |
| **July** | | | |
| **August** | | | |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total** | **74,735** | **69,207** | **-7.4%** |

***Important note**: Factors used for unit conversion: Electricity (0.0036) and Natural Gas (0.0347)*

## 3.4 WATER CONSUMPTION

According to Table 30, the actual water consumption and associated costs in February 2024 has increased by 9.1% and 9.0% respectively compared to the same month in 2023 respectively. To date, MGM COTAI has changed both the actual water consumption and associated cost by 13.7% and 13.7% respectively.

*Table 30. COTAI - Summary of Actual Monthly Water Consumption and Cost*

| Monthly Period | Actual Water Consumption ($m^3$) | | | Actual Water Cost (MOP) | | | Normalization Factors | |
|---|---|---|---|---|---|---|---|---|
| | 2023 | 2024 | % Change Actual | 2023 | 2024 | % Change Actual Cost | % Change – CDD | % Change – Food Covers |
| **January** | 54,018 | 64,264 | +19.0% | 419,790 | 499,180 | +18.9% | +173.7% | +49.1% |
| **February** | 61,371 | 66,870 | +9.1% | 476,420 | 519,440 | +9.0% | +17.2% | +48.7% |
| **March** | | | | | | | | |
| **April** | | | | | | | | |
| **May** | | | | | | | | |
| **June** | | | | | | | | |
| **July** | | | | | | | | |
| **August** | | | | | | | | |
| **September** | | | | | | | | |
| **October** | | | | | | | | |
| **November** | | | | | | | | |
| **December** | | | | | | | | |
| **Total** | **115,335** | **131,134** | **+13.7%** | **896,210** | **1,018,620** | **+13.7%** | **+62.7%** | **+48.9%** |

***Important note:*** *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

NV5 summarized the key items which could result in a water consumption increase below:

- Kitchen water consumption for February 2024 has increased by 22.6% compared with the same month in 2023.
- Mansion water consumption for February 2024 has increased by 37.4% compared with the same month in 2023.

Table 31 presents the comparison of water rates in the years 2023 and 2024. The water rate in February 2024 has decreased by 0.02% compared to the same month in 2023. The average utility rate of water the average utility rate of water has decreased by 0.04% compared to 2023.

*Table 31. COTAI - Summary of Actual Monthly Water Rates*

| Monthly Period | Year 2023 – Water Rate (MOP/m³) | Year 2024 – Water Rate (MOP/m³) | % Change – Water Rate |
|---|---|---|---|
| **January** | 7.771 | 7.768 | -0.05% |
| **February** | 7.770 | 7.768 | -0.02% |
| **March** | | | |
| **April** | | | |
| **May** | | | |
| **June** | | | |
| **July** | | | |
| **August** | | | |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Average** | **7.771** | **7.768** | **-0.04%** |

Table 32 summarized the monthly normalized water consumption. The actual water consumption in February 2024 has increased by 9.1% compared to the same month in 2019. To date, the average utility rate of water has increased by 7.1% compared to 2019.

*Table 32. COTAI - Summary of Normalized Monthly Water Consumption*

| Monthly Period | 2019 Normalized Water Consumption (m³) | 2024 Actual Water Consumption (m³) | % Change – Normalized |
|---|---|---|---|
| **January** | 61,128 | 64,264 | +5.1% |
| **February** | 61,304 | 66,870 | +9.1% |
| **March** | | | |
| **April** | | | |
| **May** | | | |
| **June** | | | |
| **July** | | | |
| **August** | | | |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total** | **122,432** | **131,134** | **+7.1%** |

## 3.5 COTAI YEAR 2024 PROJECTS

NV5 will continue to perform analysis and coordinate with MGM to implement the following projects in 2024 and reduce the site energy and water consumption which supports MGM energy and water savings target.

*Table 33. COTAI Year 2024 Projects*

| Item | Initiatives | Status |
|---|---|---|
| **Electricity** | | |
| 1 | Chiller Plant Controls Optimization<br>(CHWP DP Setpoint Reset, CHWST Setpoint Reset…) | On-going |
| 2 | PICCV Implementation 2024 (14 Units) | On-going |
| 3 | AHU EC Fan Replacement (Phase 1) | On-going |
| 4 | Additional VSD for Exhaust Air Fan (Phase 2 : 12 units) | On-going |
| 5 | KEF Steps Speed Control Optimization 2024 (Hu Dao, Coast, Grill 58, Mansion) | On-going |
| 6 | Back of House Offices Lighting Replacement | On-going |
| **Water** | | |
| 7 | Extension of B2 Carpark CDP Water Recycling (Phase 11) | Completed |
| 8 | Extension of B2 Carpark CDP Water Recycling (Phase 12) | On-going |

# APPENDIX A

## Regression Analysis

The 2019 normalized consumption modelling is remained used for tracking the agreed normalized energy savings target for MGM MACAU and MGM COTAI. The following tables show a summary of the selected independent variables for each utility which are used for the normalized consumption modelling in 2024.

## MGM MACAU

| Utilities | Variables | |
|---|---|---|
| **Electricity (kWh)** | CDD (Base T=17°C) & Food Cover | |
| **LPG (kg) / Natural Gas ($m^3$)** | Min OAT | |
| **Water ($m^3$)** | CDD (Base T=17°C) | Food Cover |

## MGM COTAI

| Utilities | Variables | |
|---|---|---|
| **Electricity (kWh)** | CDD (Base T=17°C), Food Cover, Room Night, E-Bus Mileage | |
| **Natural Gas ($m^3$)** | Min OAT | |
| **Water ($m^3$)** | CDD (Base T=17°C) | Food Cover |

# APPENDIX B

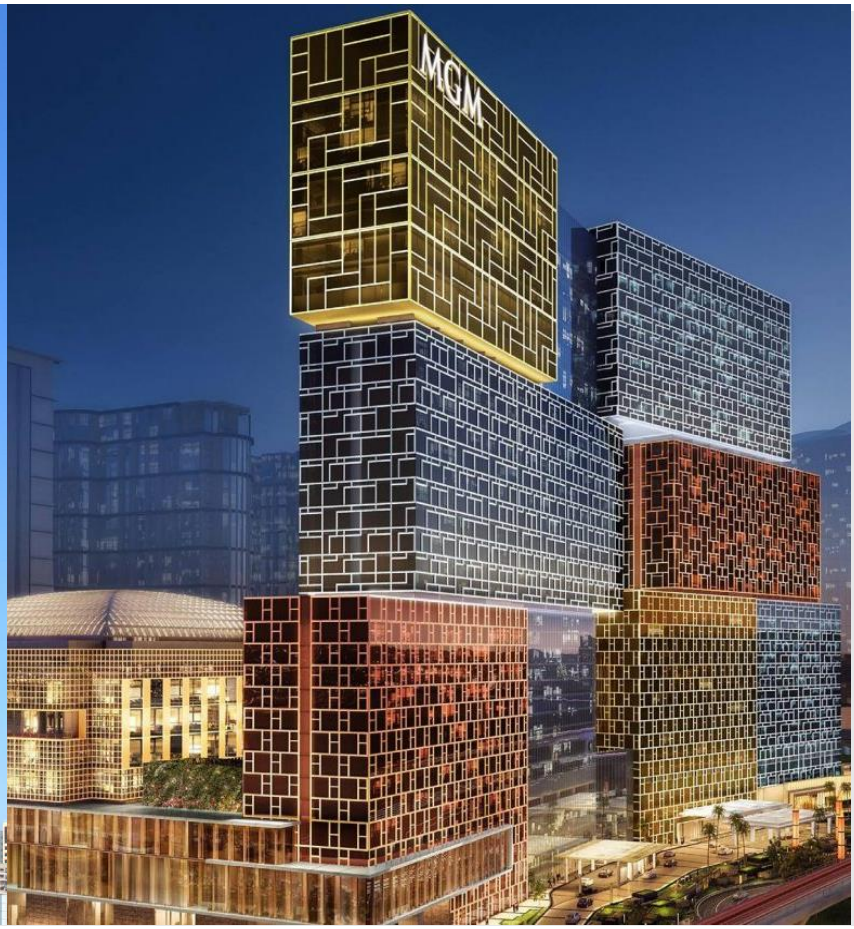
## MACAU Gas Consumption Raw Data

*Table 34. MACAU Gas Consumption Raw Data*

| Monthly Period | 2023 LPG Consumption (kg) | 2024 LPG Consumption (kg) | 2024 NG Consumption ($m^3$) |
|---|---|---|---|
| **January** | 123,108 | 50,003 | 100,712 |
| **February** | 78,483 | - | 158,122 |
| **March** | 77,597 | - | |
| **April** | 77,139 | - | |
| **May** | 80,383 | - | |
| **June** | 71,232 | - | |
| **July** | 67,441 | - | |
| **August** | 71,984 | - | |
| **September** | 69,953 | - | |
| **October** | 78,004 | - | |
| **November** | 77,482 | - | |
| **December** | 117,003 | - | |
| **Total** | **989,808** | **50,003** | **258,834** |

# MGM MACAU & COTAI
## Resource Efficiency Management Report
860723-0004415.00
860723-0004367.00




NV5 NET ZERO

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